

Dynamic Analysis of Composite conical sandwich shell with laminated Carbon epoxy face sheets

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by

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This is to certify that the work presented in the thesis entitled “**Dynamic Analysis of Composite conical sandwich shell with laminated Carbon epoxy face sheets**” by *Md Yusuful Haque*, Roll Number 122ME0205, is a record of original research carried out by him under my supervision and guidance in partial fulfilment of the requirements of the degree of *Bachelors of Technology in Mechanical Engineering*. Neither this thesis nor any part of it has been submitted for any degree or diploma to any institute or university in India or abroad.

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Dedicated
To
My Parents

Declaration of Originality

I, *Md Yusuful Haque*, Roll Number 122ME0205 hereby declare that this thesis entitled **“Dynamic Analysis of Composite truncated conical sandwich shell with laminated Carbon epoxy face sheets”** presents my original work carried out as a Bachelors student of NIT Rourkela and, to the best of my knowledge, contains no material previously published or written by another person, nor any material presented by me for the award of any degree or diploma of NIT Rourkela or any other institution. Works of other authors cited in this thesis have been duly acknowledged under the sections “Reference”.

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Abstract

Truncated conical sandwich shells are used in various modern applications such as in rocket launch vehicles, naval and automobile propulsion shafts. This study investigates dynamic behaviour of truncated conical sandwich shell of laminated carbon epoxy composite face sheets and soft epoxy core. Three finite element approaches are proposed: (i) equivalent non-uniform beam ring element modelling, where the elastic modulus is computed for the laminated face layers and (ii) axisymmetric 2-D shell modelling (iii) 3-D solid element modelling approach. The natural frequencies and mode shapes are obtained for different boundary conditions and fibre orientations in face layers. Also, the thickness of the core layer is varied and its relative effect on natural frequencies is also investigated. The results demonstrate that the epoxy core effectively raises structural stiffness while retaining lightweight characteristics, and that the tapered geometry strongly influences modal behaviour. Finally, the model is fabricated using 3-D printed conical resin blocks. The results are shown as figures, graphs and tables.

Keywords: Carbon Fiber Reinforced Polymer (CFRP) skin, Cross-ply laminate, Epoxy core, Finite element modeling, Natural frequencies, Sandwich composite structure, Truncated conical shell.

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Chapter 1

Introduction

Recently there is growing interest in lightweight yet strong structural components in launch vehicles, aerospace, marine and automobile applications. Although traditional metal structure components were well tested and efficient, they lack sufficient strength to weight ratio required for modern applications such as rocket casing, drive shaft, adaptor, pressure shells etc. Therefore, Carbon Fiber Reinforced Polymer (CFRP) composites find wide applications due to high stiffness to weight ratio, corrosion resistance and higher fatigue strength [20].

Recently, composite sandwich conical shells are frequently employed in aerospace components, submarine hulls, and architectural shells, where vibrational behavior plays a crucial role in their performance and structural integrity. Understanding and optimizing the dynamic characteristics of these shells can significantly improve their operational efficiency and safety [9], [10]. Such variable thickness structures have improved mechanical behavior, which includes enhancing stiffness, minimizing stress concentration, lowering centrifugal stress and reducing mass.

Truncated conical shells are one of the most frequently utilized structures in many industries such as petrochemical, refinery, power plant, defense, aerospace, food and so on. Since most of these types of shells cannot be included in the category of thick structural elements, the natural frequencies and buckling loads of the shell are reduced and resonance may even occur when only a small amount of load is applied. Also, significant deformations may occur under a light load. On the other hand, thick shells need more materials, thus they have more weight and higher manufacturing costs. As an effective method to overcome these disadvantages, the use of sandwich shells is a good alternative [1]. Such technique has been employed in launch rockets, storage tanks and hoppers to transfer materials.

Advanced versions of these structures can be the core-filled composite shell structure. Here, two CFRP shells are mounted one inside the other and the space in between them is filled with tough core materials such as metals and thermoplastic resins. This sandwich structure behaves like an impact resistant structure where the CFRP skin shells help in carrying the loads, while the epoxy core carries the shear and also enhances the overall

bending stiffness and load carrying capacity of the structure [3], [4]. In case of tapered shape of this structure, like a truncated cone, the behavior becomes somewhat more complicated than in case of uniform shells due to varying radius of the structure along the length, which also affects the stiffness and distribution of stress throughout the body [11].

In case of straight cylindrical shells, the radius is constant, thus the analysis becomes easier compared to tapered shells whose radius varies along the length, affecting vibration, buckling and impact performance [5]. Besides that, CFRP is anisotropic in nature and along with its varying shape it becomes more complex for analysis compared to the theories for cylindrical shells [21]. Therefore, modelling and analyzing the actual behavior becomes essential using analysis and experimentation methods.

It is must to understand that dynamic response of these structures as they may encounter vibration, axial load, bending and impact loads throughout the service life. The natural frequency and mode shape are also dependent on variables such as taper angle, CFRP layer thickness, core thickness and fiber orientation of composite layers [19]. The epoxy core is beneficial in terms of increasing the moment of inertia as well as damping and also prevents local buckling of thin layers [14], [18].

The work here is focused on the analysis of epoxy core filled tapered CFRP cylinders under various mechanical loads like impact, bending and vibration using experimental methods along with ANSYS and MATLAB models.

1.1. Background

The CFRP composite cylinders and conical shells are widely studied with respect to vibration, buckling and impact resistance [7], [11], [17], whereas sandwich shells with either foam or honeycomb core have also gained prominence due to their good stiffness and energy absorbing capability [3], [4], [5]. However, composite cylinders with solid epoxy core and two CFRP layers have a unique application in which the CFRP carries the maximum part of the load whereas the epoxy core carries shear and prevents local buckling [14], while the tapered shape makes the analysis more complicated [2], [19].

Thus, the behaviour cannot be simply expressed using conventional shell theory or sandwich shell theory because the radius varies along the length of the structure [6], [11]. In launch vehicles, a combination of hollow cylindrical and truncated conical shell structures is frequently found [2], [19]. The strength and temperature resistance can be enhanced by using sandwich structures [9]. A wide variety of materials are used in such structures [8], [18].

1.2 Motivation

Based on the importance of truncated conical shells (hollow) in various engineering applications, they are employed with different material combinations.

In this line, over the past one decade, many studies have been concentrated on

- Straight cylindrical CFRP shells,
- Sandwich shell having foam or honeycomb core material,
- Analytical and numerical studies with experimentation on tapered core filled shells.

The studies focused on epoxy core filled tapered CFRP structures motivated the present work. They help in development of high impact resistant launching systems and relatively higher tensile and shear properties. Many studies motivated the present work include:

- Light weight structures with improved stiffness and damping,
- Structures having conical profile (practical),
- Combination of analytical and experimental results for both dynamic and static properties of the structure,
- Experimentations required for better understanding the static and dynamic behaviour of the tapered core filled composite shells.

1.3 Scope of Present Work

There are many open areas on the analysis of composite sandwich conical shells with polymeric cores including mechanical characterization, thermal analysis, impact resistance, buckling stability etc. However, this study is on the analysis of conical CFRP composite sandwich shell where its bending and free vibration characteristics are the main focus. The modelling of truncated cone geometry of CFRP composite shell, equivalent composite material properties estimation for the composite-epoxy section, modelling of vibration and buckling behaviour and its analysis using Finite Element Analysis, the performance analysis under bending load, impact load and testing for mechanical hardness of composite sections can be investigated.

1.4 Thesis Outline

The structure of the thesis is organized in the following pattern:

- Chapter 1 provides introduction, motivation, scope and organization of the work.
- Chapter 2 contains a literature survey to recognize the research gap and objective of present work.
- Chapter 3 deals with geometric modelling of the structure, material modelling of the composite sandwich structure, governing equation formulation, its solution methodology and experimental setup for all kind of test, etc.

- Chapter 4 is dedicated to present and discuss the various results obtained using MATLAB code, ANSYS and experiments.
- Chapter 5 deals with the conclusions of the work and directions for future work.

Chapter 2

Literature Review

This chapter presents detailed literature review on different issues on conical sandwich shells.

2.1 Sandwich Composite Shell Structures

Fig.2.1 shows the truncated conical shell structure used in several applications.

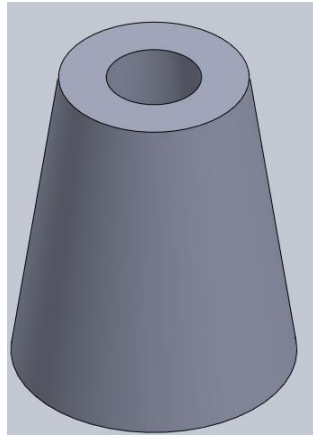


Fig.2.1 Truncated conical shell

In order to achieve high performance lightweight shells, often composite materials are used. Use of composites as sandwich structural materials is more common in modern applications, offering superior specific stiffness, strength, and damage resistance over monolithic materials [1]. Sofiyev [2] presented a state-of-the art literature review on the vibration and buckling studies of functionally graded material (FGM) composite conical shells.

In general, a sandwich structure is composed of two thin, stiff face sheets separated by a relatively thick, lightweight core material. The face sheets bear most of the in-plane and bending stresses while the core supports the shear load and prevents face sheet buckling and wrinkling [3]. Since the development of sandwich construction, there has been significant evolution in their design, initially featuring honeycomb cores, made of Aluminium and Aramid fibre reinforced composites. Subsequently, variety of other cores such as foam, corrugated, lattice and solid polymer core, have been explored. It has been proved that with polyurethane foam as filler for corrugated sandwich cylindrical panels, damping can be increased to a certain extent without any significant change in the natural frequencies, as presented by Yang *et al.* [4].

Core layer flexibility can play an important role in the sandwich response. Higher order sandwich panel theory accounting for transverse flexibility of core has been proposed,

which has also shown that soft core result in nonlinear displacement patterns through thickness [5]. This proved the inability of classic equivalent single layer theories to simulate local effects and relative displacements between the two face sheets.

Higher order sandwich panel theory for cylindrical shells was developed by Rahmani *et al.*[6] to analyse sandwich cylindrical shells with flexible cores and showed that sandwich cylindrical shells with flexible cores have much more complex vibration patterns than a homogeneous cylinder due to relative motion between the two face sheets, that results in symmetric and anti-symmetric modes.

Pourmoayed *et al.*[7] conducted vibration analysis of cylindrical sandwich panel with flexible core. Higher order shell theory was employed.

The applications of polymer core sandwich structures have also developed considerably over the last decade and due to easy manufacturing and additional functionality; these are gaining importance in various applications. Review of the use of polylactic acid (PLA) as core material in composite sandwich structures has been published and its biodegradability, recyclability and good mechanical properties mentioned by Farah *et al.* [8].

In particular, study of influence of temperature and porosity on free vibration response of the sandwich structures with 3D printed PLA cores conducted by Kallannavar and Kattimani [9] shown a great reduction in the natural frequency values with the increase in the values of both parameters.

2.2 Static and Dynamic Analysis of Composite conical sandwich Shells

Dynamic behaviour of composite shells has been widely analysed with analytical, numerical and experimental methods. It is important to identify vibration characteristics, buckling and impact response in composite structures.

Evolution from classical plate theory to higher order analyses for the composite shells was found in literature. Natural frequency calculation by Rayleigh-Ritz method on CFRP sandwich cylindrical shells, with third-order shear deformation theory and experimental verification was presented by Liu and Chen [10].

The free vibration of laminated composite and sandwich shells has also been solved using higher order closed-form solutions and dependence on geometric and material parameters was indicated by Garg *et al.*[11]. Damping behaviour in composite structures have also been analysed with the help of the dynamic analysis where for viscoelastic cores, temperature and damping were significant influencing parameters as reported by Meunier and Shenoï [12]. Malekzadeh *et al.* [13] illustrated the dynamic response of composite sandwich panels under low velocity impact. Frostig [14] presented buckling analysis of sandwich panel

with flexible core where classical buckling formulas were indicated to overestimate the critical load as it neglected the core flexibility.

Numerical analysis for composite shells has widely employed finite element software such as ANSYS and ABAQUS. The finite element prediction has also been experimentally validated by carrying out modal analysis. Arvind Raj [15] conducted modal analysis of composite sandwich structure with PLA core.

2.3 Composite Face Layer Shells and CFRP Face Sheets in Shell Structures

Carbon Fiber Reinforced Polymer (CFRP) composites are commonly utilized in shell structures due to their high specific stiffness, corrosion resistance, and excellent fatigue strength. Damping behaviour in CFRP structures have been observed to be influenced by the composite matrix itself, as well as by the sandwich core as reported by Chandra et al.[16]. Yang et al. [17] used numerical analyses to determine the modal response of sandwich cylindrical shells with CFRP face sheets showing how the natural frequencies can be significantly influenced by both fibre orientation and the face sheet thickness. The application of CFRP composite in structures like aircraft components has highlighted their specific stiffness advantage over monolithic materials as reported by Soutis[18].

Maraghi [19] investigated frequency analysis of a 3-layer sandwich truncated conical shell with FG porous core and carbon nanotube reinforced composite face sheets using first order shear deformation theory.

Wang et al. [20] studied the flutter behaviour of truncated porous sandwich conical shell with simply supported boundary conditions under supersonic wind flow. The structure has two carbon fibre surface sheets and aluminium foam core.

Shao et al.[21] obtained the free and forced vibration behaviour of FGM sandwich conical shell with different boundary conditions.

2.4 Identified Research Gaps

Based on the above literature review, following gaps are identified.

- Available research primarily focuses on the application of foam or honeycomb as core for the sandwich structures; solid epoxy as core material for the sandwich cylindrical shells is scarcely discussed.
- Studies involving tapered geometries and core filled CFRP shells are limited.
- Experimental verification for the tapered core filled composite shells is scarcely reported.
- Influence of fibre orientation on the dynamics behaviour of the mentioned structures have not been thoroughly studied.

- Lack of research comparatively analysing the performance of hollow and filled shell structures.

2.5 Objectives of the Present Study

Configuration of the shell structure considered is formed by two CFRP shells and a core between them. This would increase the bending stiffness of the structure and the impact resistance. In addition, a shell of conical shape will significantly leads to improvement in the buckling stability as well. Following are the objectives of the present work:

1. Perform the modelling of the conical shell system with CFRP face layers and epoxy core using selected geometry and material properties.
2. To develop MATLAB code to predict natural frequencies.
3. To carry out a comprehensive finite element analysis using ANSYS for vibration analysis.
4. To study the effects of taper angle, CFRP thickness, core thickness and fibre orientation.
5. To fabricate tapered composite cylindrical shells with concentric CFRP layers, filled with epoxy (ratio 10:1).

Chapter 3

Theoretical and Mathematical Modelling

This chapter presents detailed modelling of conical sandwich shells with solution approaches.

3.1 Description of Composite shell

3.2 Material Modelling and Equivalent Property Formulation

3.3 Governing Equations for Dynamic Analysis

3.4 Solution Methodology

3.5 Experimental Methodology: Fabrication Procedure

3.1 Geometry of Tapered (Truncated Cone) Composite Cylinders

Following assumptions are considered:

1. There is no slip between the layers
2. Epoxy core layer is isotropic
3. Vibration amplitudes are small compared to dimensions of the shell
4. Thickness of the shell is small relative to the radius.

Fig.3.1 shows the geometrical configuration of truncated conical shell of thickness h , lower radius a and upper radius b with length L . There are 3 layers: inner layer of thickness h_f , middle or core layer of thickness h_c and outer layer of thickness h_f .

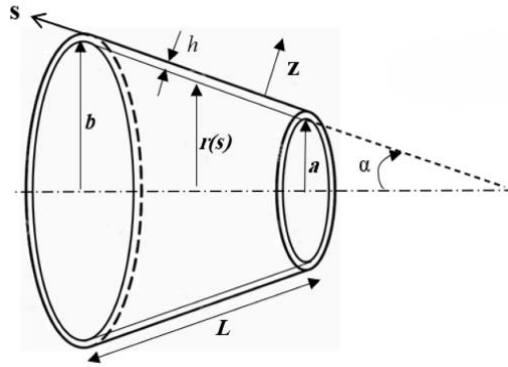


Fig 3.1 Configuration of the conical shell ($r(s)=a+s \sin\alpha$)

The displacement field of the shell in each layer is described by

$$\begin{aligned} u(s, \theta, z) &= u_0(s, \theta) + z \phi_s(s, \theta), \\ v(s, \theta, z) &= v_0(s, \theta) + z \phi_\theta(s, \theta) \\ w(s, \theta, z) &= w_0(s, \theta) \end{aligned} \quad (3.1)$$

where $u_0(s, \theta)$, $v_0(s, \theta)$, and $w_0(s, \theta)$ are the mid-surface displacements in each layer in s , θ and z directions. $\phi_s(s, \theta)$ and $\phi_\theta(s, \theta)$ represent the rotations of the normal to the mid-surface about the circumferential and meridional axes, respectively.

In the case of a conical shell, the strain-displacement relations are derived as follows

$$\epsilon_{ss} = \frac{\partial u}{\partial s} + z \frac{\partial \phi_s}{\partial s} \quad (3.2)$$

$$\epsilon_{\theta\theta} = \frac{1}{r} (u \sin \alpha + \frac{\partial v}{\partial \theta} + w \cos \alpha) + z (\phi_s \sin \alpha + \frac{\partial \phi_\theta}{\partial \theta}) \quad (3.3)$$

$$\gamma_{\theta z} = k \left(-\frac{1}{r} v \cos \alpha + \frac{1}{r} \frac{\partial w}{\partial \theta} + \phi_\theta \right) \quad (3.4)$$

$$\gamma_{sz} = \kappa \left(\frac{\partial w}{\partial s} + \phi_s \right) \quad (3.5)$$

$$\gamma_{s\theta} = \frac{1}{r} \frac{\partial u}{\partial \theta} + \frac{\partial v}{\partial s} - \frac{1}{r} v \sin \alpha + z \left(\frac{1}{r} \frac{\partial \phi_s}{\partial \theta} + \frac{\partial \phi_\theta}{\partial \theta} - \frac{\sin \alpha}{r} \phi_\theta \right) \quad (3.6)$$

- α is the semi-vertex angle of the cone.

- κ is the shear correction factor, typically taken as $\kappa = 5/6$ for homogeneous isotropic materials.

Stress-Strain Relationship for Conical Shell

$$\sigma_{ss} = Q_{11}\epsilon_{ss} + Q_{12}\epsilon_{\theta\theta} \quad (3.7)$$

$$\sigma_{\theta\theta} = Q_{12}\epsilon_{ss} + Q_{22}\epsilon_{\theta\theta} \quad (3.8)$$

$$\tau_{s\theta} = Q_{66}\gamma_{s\theta} \quad (3.9)$$

$$\tau_{sz} = Q_{44}\gamma_{sz} \quad (3.10)$$

$$\tau_{\theta z} = Q_{55}\gamma_{\theta z} \quad (3.11)$$

Where Q_{ij} are material stiffness coefficient which depend on material properties of conical shell. Here,

$$Q_{11} = \frac{E_{11}}{1-\gamma_{12}\gamma_{21}}, Q_{22} = \frac{E_{22}}{1-\gamma_{12}\gamma_{21}}, Q_{12} = \frac{\gamma_{12}E_{22}}{1-\gamma_{12}\gamma_{21}}, Q_{44} = G_{23}, Q_{55} = G_{13} \text{ and } Q_{66} = G_{12} \quad (3.12)$$

With E_{11} , E_{22} , ν_{12} , ν_{21} , G_{12} etc. as elastic modulus, Poisson ratio and shear modulus of skin.

3.2 Material Modelling and Equivalent Property Formulation

For the face layers, the resultant orthotropic properties are given as:

E_1 = longitudinal modulus

E_2 = transverse modulus

G_{12} = in-plane shear modulus

ν_{12} = major Poisson ratio

If θ = ply angle and N are number of plies, the axial and bending stiffness of the skin are given as:

Axial and Bending stiffness

When the conical shell is considered as beam, the axial stiffness is given by

$$A_{11} = \sum_{k=1}^n (\overline{Q_{11}})_k (z_k - z_{k-1}) \quad (3.13)$$

And bending stiffness is

$$D_{11} = \sum_{k=1}^n (\overline{Q_{11}})_k (z_k^3 - z_{k-1}^3) \quad (3.14)$$

$\overline{Q}_{11}$ is transformed stiffness of layer k given as:

$$\overline{Q}_{11} = Q_{11}c^4 + Q_{22}s^4 + 2(Q_{12} + Q_{66})s^2c^2 \quad (3.15)$$

Where

$$Q_{11} = \frac{E_1}{1-\nu_{12}\nu_{21}} \quad (3.16a)$$

$$Q_{22} = \frac{E_2}{1-\nu_{12}\nu_{21}} \quad (3.16b)$$

$$Q_{12} = \frac{\nu_{12}E_2}{1-\nu_{12}\nu_{21}} \quad (3.16c)$$

$$Q_{66} = G_{12} \quad (3.16d)$$

Also, c and s are cos and sin of the angle. The longitudinal and transverse elastic modulus are then given as:

$$E_{axial} = A_{11}h \quad (3.17)$$

$$E_{bending} = \frac{12 D_{11}}{h^3} \quad (3.18)$$

Where h is the total thickness of layer.

3.3 Governing Equations for Dynamic Analysis

Based on Hamilton's principle, the governing equations are derived for an arbitrary time interval using the variation operator δ as

$$\int_{t_1}^{t_2} (\delta K - \delta U) dt = 0 \quad (3.19)$$

where, K corresponds to the kinetic energy of the shell, while U represents the strain energy.

Kinetic energy:

The kinetic energy of the shell is determined by the following equation:

$$K = \frac{1}{2} \int_V \rho \left[\left(\frac{\partial u}{\partial t} \right)^2 + \left(\frac{\partial v}{\partial t} \right)^2 + \left(\frac{\partial w}{\partial t} \right)^2 \right] dV \quad (3.20a)$$

$$K = \int_S \left(\int_{-\frac{h_c}{2}}^{\frac{h_c}{2}} \rho_o \left[\left(\frac{\partial u}{\partial t} \right)^2 + \left(\frac{\partial v}{\partial t} \right)^2 + \left(\frac{\partial w}{\partial t} \right)^2 \right] dz + \int_{-\frac{h_c}{2}}^{\frac{h_c}{2}} \rho_c \left[\left(\frac{\partial u}{\partial t} \right)^2 + \left(\frac{\partial v}{\partial t} \right)^2 + \left(\frac{\partial w}{\partial t} \right)^2 \right] dz \right. \\ \left. + \int_{\frac{h_c}{2}}^{\frac{h_c}{2}+h_o} \rho_o \left[\left(\frac{\partial u}{\partial t} \right)^2 + \left(\frac{\partial v}{\partial t} \right)^2 + \left(\frac{\partial w}{\partial t} \right)^2 \right] dz \right) dS \quad (3.20b)$$

here V and S denote the volume and the mid-surface area of the conical shell respectively in which $dS = r(s)d\theta ds$. The rotational mass terms are defined as

$$\begin{Bmatrix} I_0 \\ I_1 \\ I_2 \end{Bmatrix} = \int_{-h/2}^{+h/2} \rho(z) \begin{Bmatrix} 1 \\ z \\ z^2 \end{Bmatrix} dz \quad (3.21)$$

Strain energy:

Likewise, the strain energy of the conical shell is expressed in terms of the stress and strain tensor as follows

$$U_s = \frac{1}{2} \int_V (\sigma_{ss} \varepsilon_{ss} + \sigma_{\theta\theta} \varepsilon_{\theta\theta} + \sigma_{\theta z} \gamma_{\theta z} + \sigma_{sz} \gamma_{sz} + \sigma_{s\theta} \gamma_{s\theta}) dV \quad (3.22)$$

In which, the stress components are defined as follows:

The in-plane force, bending moment and shear force are given as:

$$\begin{Bmatrix} N_{ss} \\ N_{\theta\theta} \\ N_{s\theta} \end{Bmatrix} = \int_{-\frac{h}{2}}^{\frac{h}{2}} \begin{Bmatrix} \sigma_{ss} \\ \sigma_{\theta\theta} \\ \sigma_{s\theta} \end{Bmatrix} dz \quad (3.23a)$$

$$\begin{Bmatrix} M_{ss} \\ M_{\theta\theta} \\ M_{s\theta} \end{Bmatrix} = \int_{-\frac{h}{2}}^{\frac{h}{2}} \begin{Bmatrix} \sigma_{ss} \\ \sigma_{\theta\theta} \\ \sigma_{s\theta} \end{Bmatrix} z dz \quad (3.23b)$$

$$\begin{Bmatrix} Q_{xz} \\ Q_{\theta z} \end{Bmatrix} = \int_{-\frac{h}{2}}^{\frac{h}{2}} \begin{Bmatrix} \sigma_{sz} \\ \sigma_{\theta z} \end{Bmatrix} dz \quad (3.23c)$$

The resultant governing equations are given as:

$$\frac{\partial N_{ss}}{\partial s} + \frac{1}{r} \frac{\partial N_{s\theta}}{\partial \theta} + \frac{1}{r} (N_{ss} - N_{\theta\theta}) \sin \alpha - I_0 \frac{\partial^2 u}{\partial t^2} - I_1 \frac{\partial^2 \phi_s}{\partial t^2} = 0 \quad (3.24)$$

$$\frac{\partial N_{s\theta}}{\partial s} + \frac{1}{r} \frac{\partial N_{\theta\theta}}{\partial \theta} + \frac{2}{r} N_{s\theta} \sin \alpha + \frac{1}{r} Q_{\theta z} \cos \alpha - I_0 \frac{\partial^2 v}{\partial t^2} - I_1 \frac{\partial^2 \phi_\theta}{\partial t^2} = 0 \quad (3.25)$$

$$\frac{\partial Q_{sz}}{\partial s} + \frac{1}{r} Q_{sz} \sin \alpha - \frac{1}{r} N_{\theta\theta} \cos \alpha + \frac{1}{r} \frac{\partial Q_{\theta z}}{\partial \theta} - I_0 \frac{\partial^2 w}{\partial t^2} = 0 \quad (3.26)$$

$$\frac{\partial M_{ss}}{\partial s} + \frac{1}{r} \frac{\partial M_{s\theta}}{\partial \theta} + \frac{1}{r} (M_{ss} - M_{\theta\theta}) \sin \alpha - Q_{sz} - I_1 \frac{\partial^2 u}{\partial t^2} - I_2 \frac{\partial^2 \phi_s}{\partial t^2} = 0 \quad (3.27)$$

$$\frac{\partial M_{s\theta}}{\partial s} + \frac{1}{r} \frac{\partial M_{\theta\theta}}{\partial \theta} + \frac{2}{r} M_{s\theta} \sin \alpha - Q_{\theta z} - I_1 \frac{\partial^2 v}{\partial t^2} - I_2 \frac{\partial^2 \phi_\theta}{\partial t^2} = 0 \quad (3.28)$$

For the boundary conditions table:

$$\text{Clamped (C): } u = v = w = 0, \quad \phi_s = \phi_\theta = 0$$

$$\text{Simply – supported (S): } v = w = \phi_\theta = 0, \quad N_{ss} = M_{ss} = 0$$

3.4 Solution Methodology

Using both analytical modelling in MATLAB and numerical simulation in ANSYS, the dynamic behaviour of the epoxy core-filled CFRP truncated cone is investigated. The geometry and material

properties of the CFRP faces sheets [0/90] and epoxy core are given exactly. In the analytical approach in MATLAB, the structure is modelled as a meridional beam which is then discretized by using finite elements, and then both global stiffness and mass matrices are constructed. From the eigenvalue problem, natural frequencies were obtained with clamped-free boundary conditions.

Using 2-D axisymmetric finite element analysis with ANSYS via SHELL181 elements for composite layers and epoxy core layer, modal analysis is carried out for obtaining the natural frequencies and the transient structural analysis is conducted to investigate the response due to the impact loads.

Third solution is obtained using 3-D finite element approach of the shell without considering axisymmetric nature. This is attempted with ABAQUS modal analysis using solid elements.

3.5 Sample fabrication

The conical shell sample for testing is prepared in the following steps:

1. Two moulds of different dimensions with conical geometry are manufactured using 3D printing machine for producing epoxy core-filled CFRP sandwich truncated cone structures.
 2. The moulds external sides have been coated by Mylar film in order to produce smooth surfaces and easy de-moulding after curing.
 3. A releasing agent based on silicone has been used to prevent adhesion of the mould surface with the laminated composite layers.
 4. The carbon fibres have been trimmed according to the mould geometry. Two layers of carbon fibres have been prepared to construct a cross-ply laminate $[0^\circ/90^\circ]$ configuration.
 5. The epoxy resin has been mixed with hardener to produce the polymer matrix at a weight ratio 10:1 for CFRP face sheets.
 6. The Mylar covered mould surface has been covered with a thin layer of epoxy–hardener matrix. The carbon fibres have been laminated in $[0^\circ/90^\circ]$ and immersed in the epoxy matrix to fabricate the laminated CFRP shell.
 7. The CFRP shells have been cured until a sufficient hardening of the polymer matrix.
 8. After complete curing, both PLA moulds and Mylar films have been removed to fabricate two CFRP truncated cones shells separately.
 9. The smaller CFRP cone has been placed inside the larger one in order to produce an annular space between them.
 10. The epoxy resin has been mixed with the hardener at the weight ratio 10:1 to introduce it to fill the annular gap.
 11. The whole system has been allowed to cure until the hardening of the epoxy core.
- Finally, an epoxy core-filled CFRP sandwich truncated cone with CFRP laminated faces sheets has been prepared.

Chapter 4

Results and Discussion

4.1 Material and geometric Properties

The "Sandwich" configuration utilizes high-stiffness face layers to handle bending loads and a lightweight core (epoxy) to provide shear rigidity and maintain the moment of inertia. Table 4.1 shows the material details of the sandwich configuration.

Table 4.1:Material Specifications

Layer	Material	Function
Outer skin	Carbon Epoxy UD Prepreg	Primary load-bearing (Axial/Hoop)
Core	Epoxy Resin	Maintains geometric separation (Shear)
Inner skin	Carbon Epoxy UD Prepreg	Primary load-bearing (Axial/Hoop)

The following primary dimensions are established and used in simulation and fabrication:

- Axial Length (L): 103.7 mm
- Face Sheet Thickness (t_f): 1.0 mm (Carbon Epoxy UD Prepreg)
- Core Material: Epoxy Resin

Table 4.2 shows the geometric properties of the sandwich conical shell considered.

Table 4.2:Geometric Specifications

Parameter	Small End (Top)	Large End (Bottom)
Inner Radius (R_{in})	12 mm	37 mm
Outer Radius (R_{out})	27 mm	52 mm
Total Wall Thickness (h)	15 mm	15 mm
Net Core Thickness (hc)	13 mm	13 mm

4.1.1 Radius Variation and Taper Angle

The semi-vertex angle (α) characterizes the linear taper of the cylinder. Using the boundary radii and the axial length:

$$\tan(\alpha) = \frac{R_{large} - R_{small}}{L} = \frac{37 - 12}{103.7} \approx 0.241$$
$$\alpha \approx 13.55^\circ$$

The local radius $r(x)$ as a function of axial distance s from the small end is:
 $r(s) = 12 + 0.241s$

4.1.2 Cross-Sectional Material Architecture

The "Sandwich" configuration is maintained uniformly throughout the length of the cone, ensuring consistent moment of inertia. Fig.4.3 shows the specifications and dimensions considered.

Table 4.3: Material and Geometric Specifications

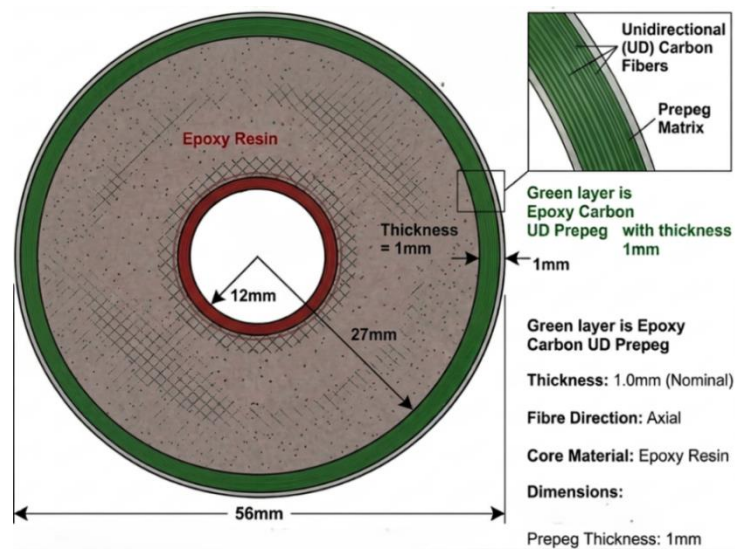
Layer	Material	Thickness	Functional Role
Outer Face	Carbon Epoxy UD Prepreg	1.0 mm	Primary Load Bearing (Axial/Hoop)
Core	Epoxy Resin	13.0 mm	Shear Transfer & Stability
Inner Face	Carbon Epoxy UD Prepreg	1.0 mm	Primary Load Bearing (Axial/Hoop)

4.1.3 Material Parameters

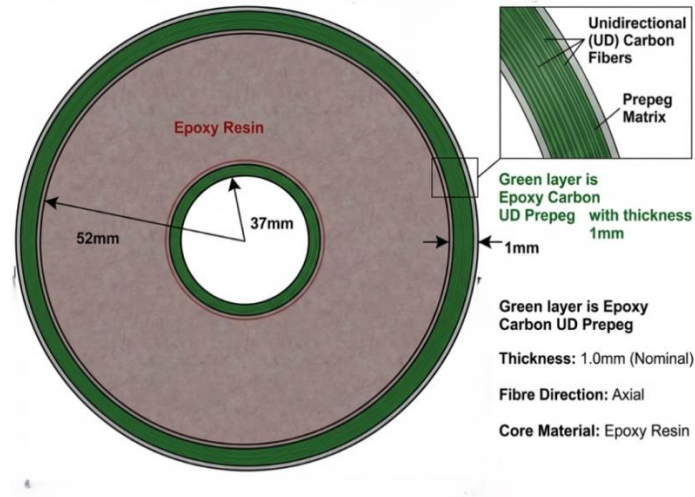
Following material properties for epoxy-carbon unidirectional (UD) prepreg are considered in 3-D modelling:

$E_x=121$ GPa, $E_y=E_z=8.9$ GPa, $\nu_{xy}=\nu_{xz}=0.27$, $\nu_{yz}=0.4$, $G_{xy}=G_{xz}=4.7$ GPa, $G_{yz}=3.1$ GPa, density considered $\rho=1490$ kg/m³. For epoxy, the isotropic properties considered are: $E=3.78$ GPa, $\nu=0.35$, density $\rho=1160$ kg/m³.

Fig.4.1 shows the cross-section of the outer and inner sandwich configurations.



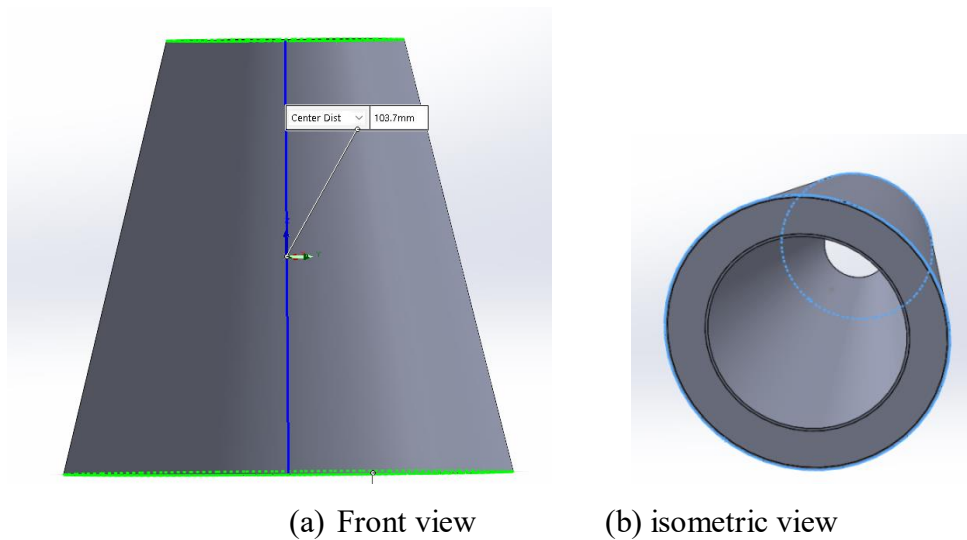
(a) Cross-Section of Inner Face Layer



(b) Cross-Section of Inner Face Layer

Fig.4.1 Geometrical details for CAD modelling

Fig.4.2 shows the CAD model developed in SolidWorks.



(a) Front view

(b) isometric view

Fig 4.2: Conical shell geometry with axial length of 103.7 mm

4.2 Effect of Core layer Thickness on Natural Frequencies

The effect of varying the thickness of epoxy core in determining the mode shapes of the CFRP sandwich truncated cone was studied using Abaqus modal analysis. Four different models were analyzed where the epoxy core thickness values were 13mm, 14mm, 15mm, and 16mm, whereas the thickness of the CFRP face sheets remained 1mm for both sides.

Table 4.4 shows the variation of natural frequencies as a function of core layer thickness. It can be seen from the results that changes in the thickness of the core have a significant effect on the dynamics of the system due to its effect on mass and stiffness.

Table 4.4 Effect of core layer thickness on the first six modes

Mode in Hz	13 mm Core	14 mm Core	15 mm Core	16 mm Core
1	2718.8	2706.6	2696.0	2686.7
2	3680.5	3623.7	3572.7	3526.7
3	5049.6	5028.9	4998.2	4963.2
4	5405.4	5485.2	5557.9	5624.3
5	5595.3	5626.6	5667.6	5712.4
6	6562.0	6527.8	6496.9	6468.8

According to the results obtained, it can be noted that the modes of lower frequencies tend to reduce with increase in core thickness as a result of increased mass structure. On the other hand, the modes with higher frequencies will increase with increase in core thickness because of increased bending stiffness. Figures 4.3 to 4.8 shows the first 6 mode shapes obtained for above frequencies corresponding to core layer thickness 13 mm.

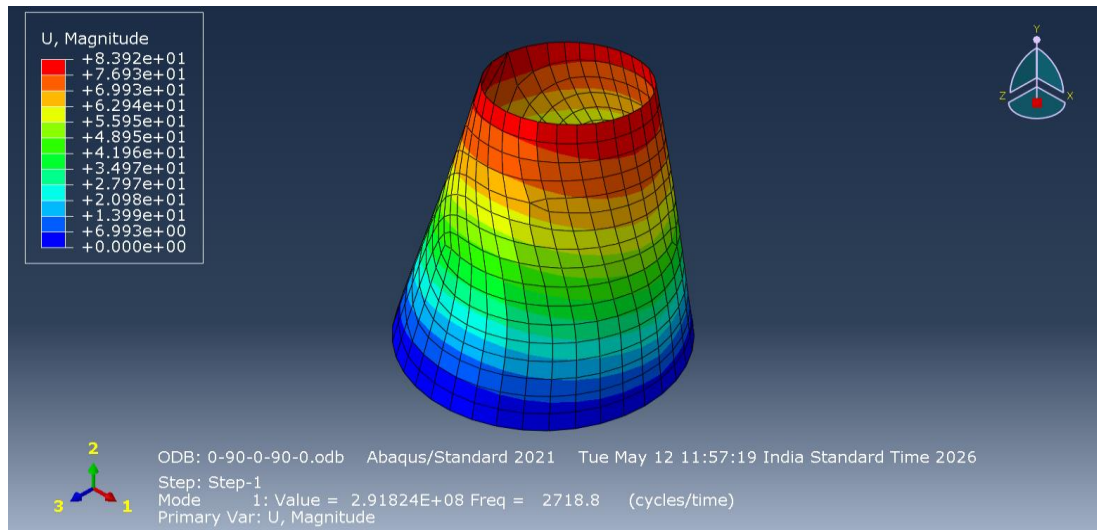


Figure 4.3: Mode Shape 1 of CFRP Sandwich Truncated Cone (2718.8 Hz)

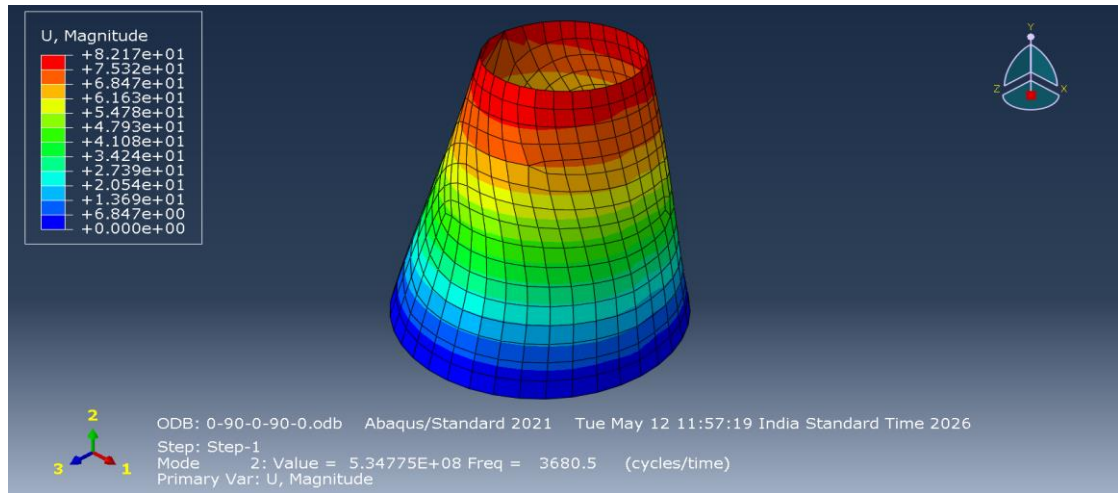


Figure 4.4: Mode Shape 2 of CFRP Sandwich Truncated Cone (3680.5 Hz)

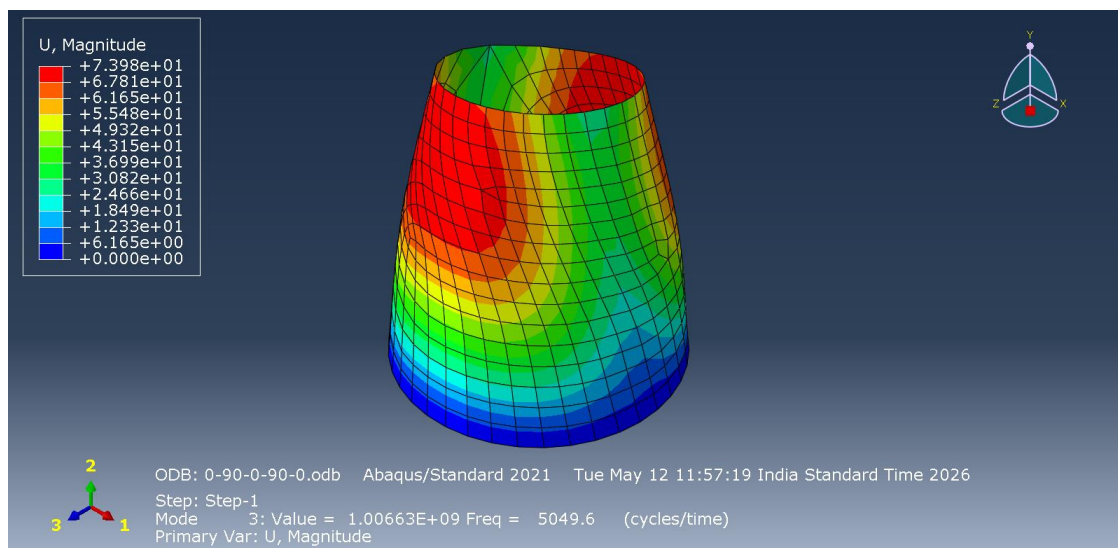


Figure 4.5: Mode Shape 3 of CFRP Sandwich Truncated Cone (5049.6 Hz)

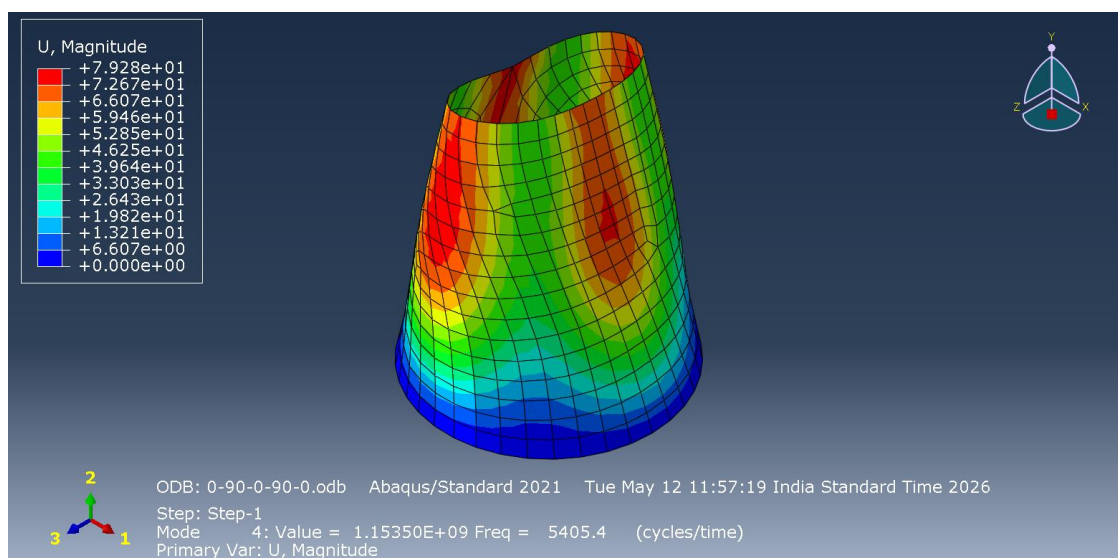


Figure 4.6: Mode Shape 4 of CFRP Sandwich Truncated Cone (5405.4 Hz)

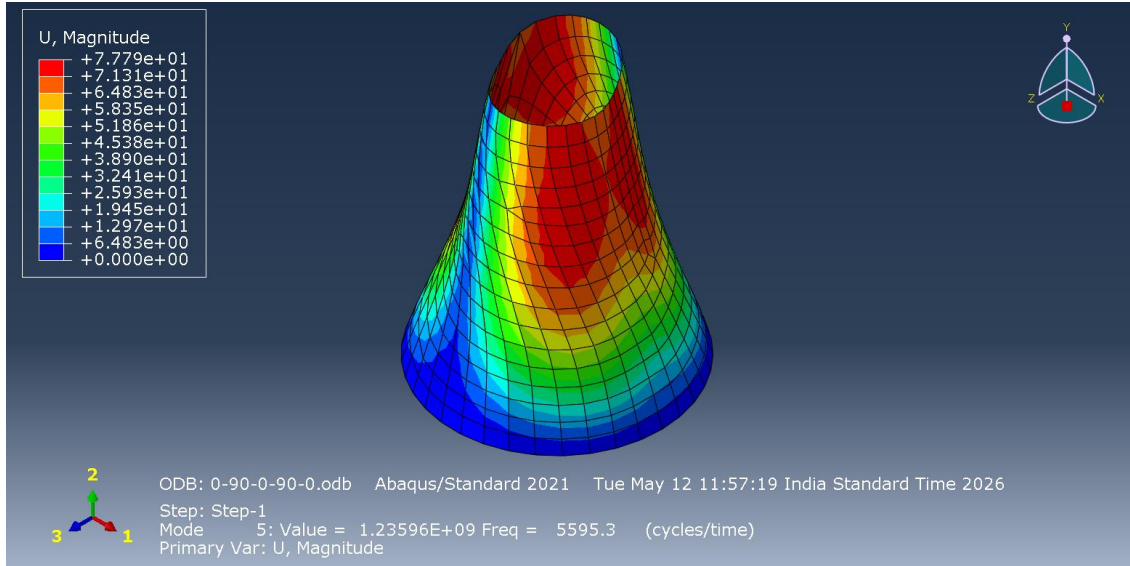


Figure 4.7: Mode Shape 5 of CFRP Sandwich Truncated Cone (5595.3 Hz)

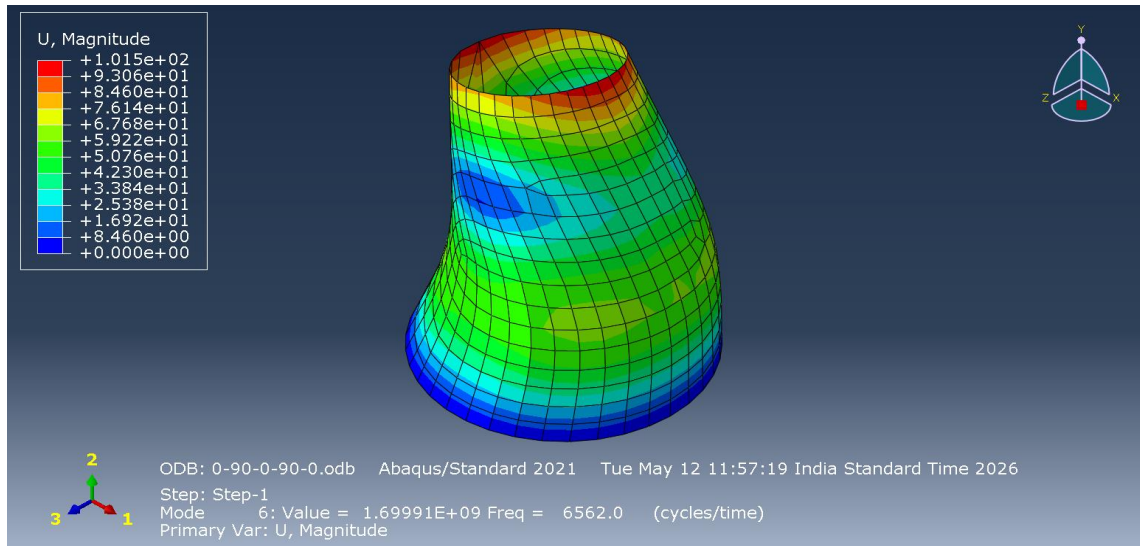


Figure 4.8: Mode Shape 6 of CFRP Sandwich Truncated Cone (6562.0 Hz)

4.3 Effect of Fiber Orientation on Natural Frequencies

The effect of laminate stacking sequence and fibre orientation on modal characteristics of the CFRP sandwich truncated cone is analysed in this investigation. The effects of different laminate orientations are analysed in terms of stiffness and modal characteristics of the structure. Table 4.5 represents natural frequencies calculated for modal analysis.

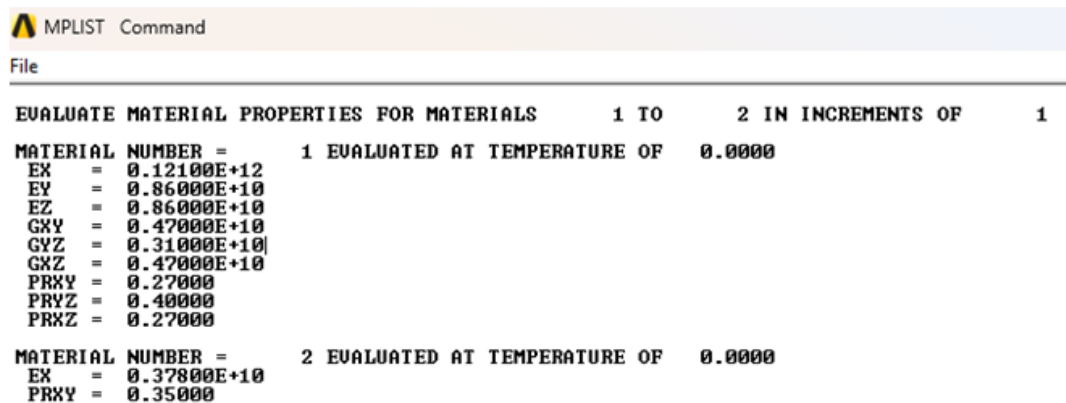
Table 4.5 Effect fibre orientation of the skin on natural frequencies

Mode	0–90 (Hz)	90–0 (Hz)	0–0 (Hz)	90–90 (Hz)
1	2718.8	2722.0	2723.6	2720.4
2	3680.5	3692.5	3690.6	3678.1
3	5049.6	5051.5	5028.6	5029.2
4	5405.4	5359.3	5295.9	5345.3
5	5595.3	5560.9	5526.7	5561.6
6	6562.0	6576.9	6584.6	6569.6

Results show that fibre orientation affects the stiffness behaviour of the laminate and consequently changes the natural frequencies and modes of vibration. The stiffer the laminate in the axial direction, the higher its modal frequencies.

4.4 Axisymmetric Finite element modelling of the shell

The conical shell is considered as a linear 2-D member in a plane with respect to its axis. This makes lesser computational work. Here, the 2-D 4-node shell elements (SHELL181) can be used to mesh the configuration. After entering following material properties (Table 4.6), the ply orientation is entered for the element. Further, the geometry of the portion is drawn in modelling window from key points to areas. Three areas representing inner and outer face layers and middle core portion are created as trapezoidal blocks.

Table 4.6 Screen shot of material data for skin and core layers


```

MPLIST Command
File
EVALUATE MATERIAL PROPERTIES FOR MATERIALS      1 TO      2 IN INCREMENTS OF      1
MATERIAL NUMBER =      1 EVALUATED AT TEMPERATURE OF      0.0000
EX  = 0.12100E+12
EY  = 0.86000E+10
EZ  = 0.86000E+10
GXY = 0.47000E+10
GYZ = 0.31000E+10
GXZ = 0.47000E+10
PRXY = 0.27000
PRYZ = 0.40000
PRXZ = 0.27000
MATERIAL NUMBER =      2 EVALUATED AT TEMPERATURE OF      0.0000
EX  = 0.37800E+10
PRXY = 0.35000

```

For analysis, the axisymmetric boundary conditions are applied such that the conical shell does not deflect at the axis radially. Fig.4.9 shows the meshed model screen shot.

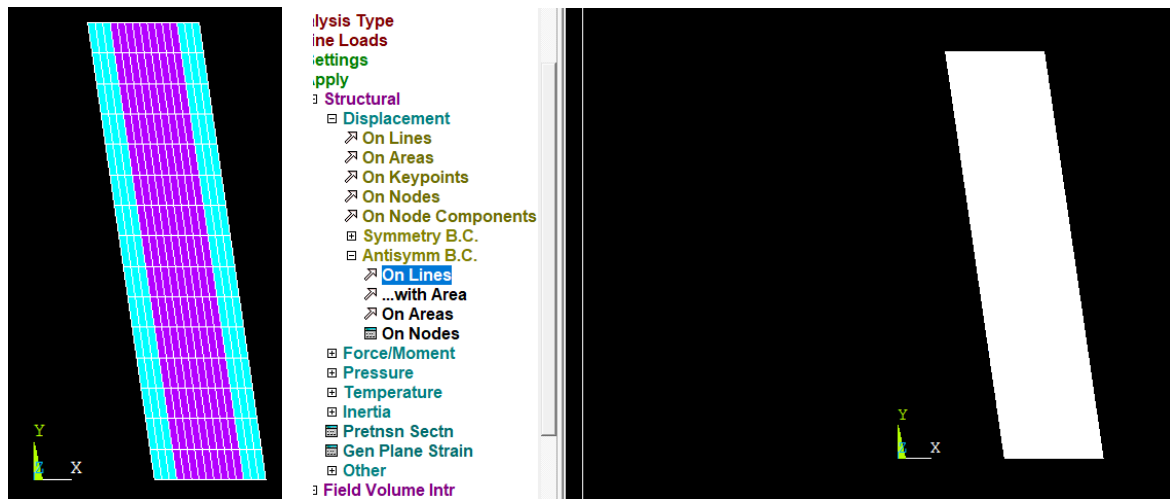


Fig.4.9 Axisymmetric model of conical sandwich shell

4.5 Beam element modelling of sandwich shell

The truncated conical shell can be considered as a tapered beam if asymmetry is considered in the system. Here, there are 2 nodes for every element and the degrees of freedom per node considered are 3 namely, u , v and θ (axial and bending). The element stiffness and mass matrices are of size 6×6 . A simple MATLAB program (See Appendix) is used to obtain the global matrices and the natural frequencies are reported. For the selected dimensions, the natural frequencies obtained are: Table 4.7 shows the first 5 natural frequencies obtained for fixed-free boundary conditions from MATLAB and Fig.4.10 shows the plot of same. Several parametric studies can be conducted by varying length, diameters of the cone and natural frequencies can be obtained. Similarly, by varying the ply orientations the frequency variation can be recorded.

Table 4.7 First Five Natural Frequencies Obtained from MATLAB Beam FEM Analysis

Mode Number	Natural Frequency (Hz)
Mode 1	802
Mode 2	5993
Mode 3	6418
Mode 4	17674
Mode 5	22646

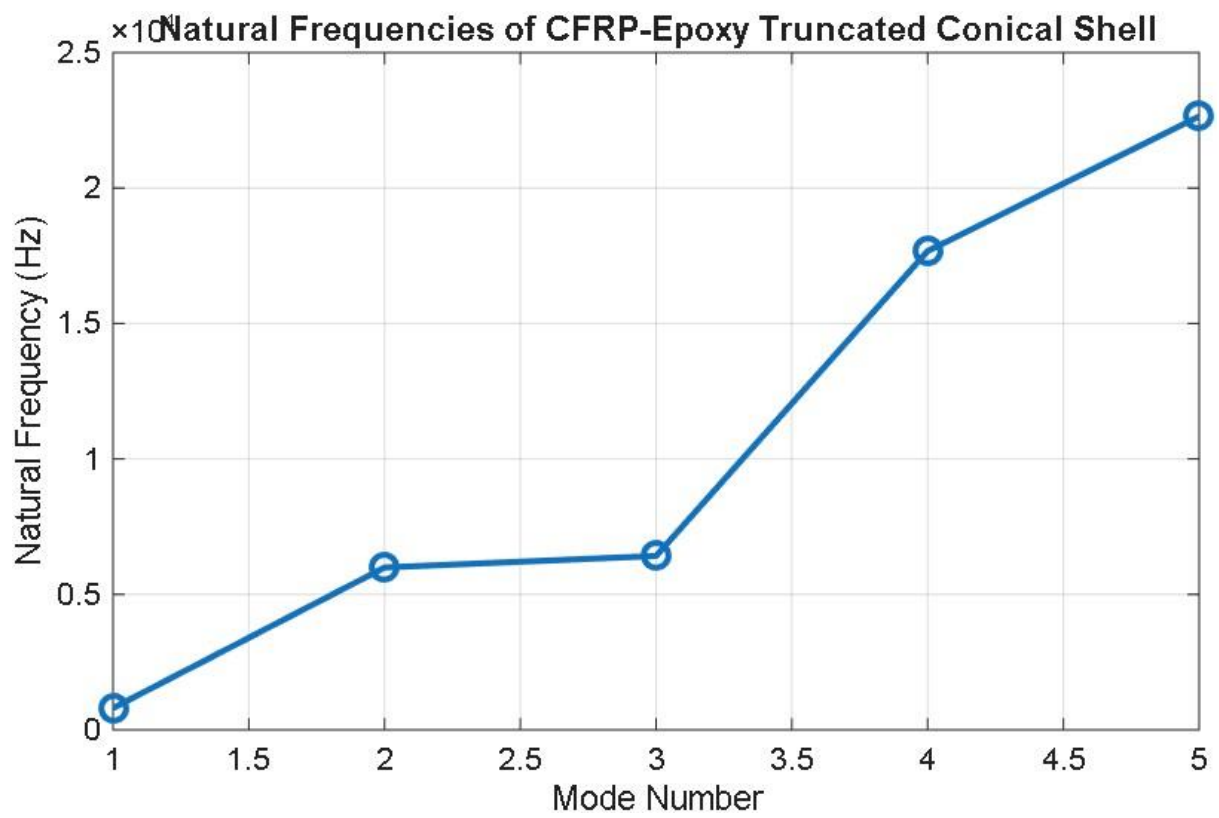


Fig. 4.10 Variation of Natural Frequencies with Mode Number for CFRP–Epoxy Truncated Conical Shell

Chapter 5

CONCLUSIONS AND FUTURE SCOPE

The study examines the dynamic behaviour of sandwich panels with epoxy core filled CFRP truncated cone structure.

5.1 Things Done in Work

1. Modelling of geometry of truncated conical sandwich shell made of laminated CFRP face sheets and epoxy core.
2. Modelling of mechanical properties of orthotropic CFRP face layers and isotropic epoxy core through equivalent laminate stiffness approach.
3. Formulation of equations for axial stiffness, bending stiffness, strain-displacement relations, and vibration analysis based on laminated shell and beam idealizations.
4. Programming of FEM-based algorithm coded in MATLAB language in order to calculate global stiffness matrix and mass matrix as well as natural frequencies.
5. Dynamic modal analysis of the composite conical shell using ABAQUS and ANSYS commercial FEM software.
6. Application of axisymmetric shell model and 3D solid finite element modelling techniques for modal analysis of conical shell.
7. Study of natural frequencies and mode shapes sensitivity to changes of epoxy core thickness (13 mm, 14 mm, 15 mm, and 16 mm).
8. Effect of fibre orientation ([0/90], [90/0], [0/0], [90/90]) of laminate on modal characteristics investigation.
9. Investigation of the effect of taper angle change on modal characteristics of conical shell via beam FEM approach.
10. Obtaining and comparing six different modes shapes and natural frequencies.
11. Fabrication of composite sandwich shell specimens from 3D printed PLA Molds and prepregs, epoxy resin and hardener, Mylar film, and silicone releasing agent.
12. Assembling of the composite structure, i.e., two CFRP truncated cones assembled concentrically with annular epoxy filled core region.

5.2 Summary of Findings

1. Dynamic analysis of epoxy core-filled CFRP truncated conical sandwich shells was successfully carried out using MATLAB beam FEM formulation and finite element analysis in ABAQUS/ANSYS.

2. The influence of epoxy core thickness variation was studied for core thickness values of 13 mm, 14 mm, 15 mm, and 16 mm.
3. Lower vibration modes (Modes 1–3) showed decreasing natural frequencies with increasing core thickness. This behaviour occurred because the increase in structural mass due to thicker epoxy core dominated over the stiffness improvement.
4. Higher vibration modes (Modes 4–5) exhibited increasing frequency trends with increasing core thickness because bending stiffness became dominant in higher modes.
5. Increase in core thickness increased the second moment of inertia and bending rigidity ((EI)), thereby improving the structural bending resistance of the sandwich shell.
6. The epoxy core significantly enhanced the bending stiffness of the composite structure while maintaining lightweight characteristics.
7. CFRP face sheets primarily carried the structural loads, whereas the epoxy core provided additional stiffness and improved structural stability.
8. Variation in taper angle strongly affected the modal characteristics and dynamic behaviour of the shell structure.
9. Increase in taper angle changed the radius distribution along the axial direction, resulting in non-uniform stiffness and mass distribution throughout the shell.
10. Larger taper angles produced more complex mode shapes and altered natural frequencies due to increased geometric non-uniformity.
11. The tapered conical shell exhibited more complicated dynamic responses compared to uniform cylindrical shells because of variable geometry and stiffness distribution.
12. The obtained results agree with the fundamental principles of sandwich beam theory and shell vibration theory.
13. The study demonstrates that epoxy core-filled CFRP truncated conical sandwich shells possess improved stiffness-to-weight characteristics and are suitable for lightweight structural applications requiring enhanced dynamic performance.
14. The laminate stacking sequence significantly influenced the modal characteristics of the truncated conical shell.

15. Laminates having higher fibre orientation in the axial direction ($[[0/0]]$) exhibited higher stiffness and comparatively larger natural frequencies.
16. Laminates dominated by transverse fibre orientation ($[[90/90]]$) showed relatively lower axial stiffness and lower natural frequencies.
17. Cross-ply laminates such as $[[0/90]]$ and $[[90/0]]$ provided balanced stiffness characteristics and stable modal behaviour.
18. The variation in fibre orientation changed the equivalent laminate stiffness matrix ($[[A],[D]]$), thereby influencing the overall vibration response and mode shapes of the sandwich shell structure.

5.3 Future Recommendation

There are several aspects that can be further explored to gain deeper insight into the dynamic behaviour of composite truncated conical sandwich shells. The following can be suggested for future work:

1. Modal analysis based on experimental techniques involving impact hammer and accelerometer testing for validation of numerically obtained results and MATLAB code.
2. Transient dynamic analysis and simulation of high-velocity impact loading to explore real-life impact effects and damage propagation mechanisms.
3. Study of different fibre directions and layup orders (e.g., $[0^\circ/\pm 45^\circ/90^\circ]$) with the goal of finding optimal design parameters providing maximum rigidity or vibration damping effect.
4. Use of innovative core materials such as CNT-filled epoxy core, foam core, or honeycomb core instead of solid epoxy cores.
5. Study of buckling, fatigue, and nonlinear vibrations in order to assess long-term stability.
7. Optimization of design parameters including thickness of core material, taper angle, and layup order of laminates in order to optimize the weight and rigidity ratio.
8. Development of more accurate shell theories and improved finite element modelling

Appendix

MATLAB Code

```
% Dynamic Analysis of Truncated Conical Sandwich Shell
% Using 2-Node Beam Elements (Finite Element
Method) clear; clc;

--- 1. Geometry and Material Properties ---
L = 103.7e-3; % Length of cone (m)

R1 = (12e-3 + 27e-3)/2; % Small end mean
radius (m) R2 = (37e-3 + 52e-3)/2; %
Large end mean radius (m)

alpha = atan((R2-R1)/L); % Cone angle

nElements = 10; % Number of beam
elements nodes = nElements + 1;
s = linspace(0, L, nodes);

--- Sandwich Properties ---
h_face = 1e-3; % Thickness of one CFRP face
sheet (m) h_core = 13e-3; % Thickness of epoxy core (m)
h = 2*h_face + h_core; % Total sandwich thickness

--- CFRP Laminate Stiffness using A11 ---
A11 = 0;
alp = [0 90]; % [0/90] laminate

E11 = 121e9; % Longitudinal modulus of CFRP ply (Pa)
E22 = 8.6e9; % Transverse modulus of CFRP
ply (Pa) nu12 = 0.27;
nu21 = nu12*E22/E11; % Correct minor
Poisson ratio G12 = 4.7e9; % Shear modulus (Pa)

Q11 = E11/(1-nu12*nu21)

Q22 = E22/(1-nu12*nu21);
Q12 = nu12*E22/(1-
nu12*nu21); Q66 = G12;

% Ply interface locations for 1 mm
face sheet z = [-h_face/2, 0,
h_face/2];

for k = 1:2
    al =
    alp(
    k);
    c =
    cosd
    (al)
    ;
    ss = sind(al);

    Qbar = Q11*c^4 + Q22*ss^4 + 2*(Q12 +
    2*Q66)*ss^2*c^2; A11 = A11 + Qbar*(z(k+1)-
    z(k));
end

E_f = A11/h_face; % Equivalent axial modulus of CFRP face sheet
```

--- Core Properties ---

```
E_c = 3.78e9; % Young's modulus of epoxy core (Pa)
nu_f = 0.27; % Poisson's ratio face
nu_c = 0.35; % Poisson's ratio core
rho_f = 1490; % Density of CFRP face (kg/m^3)
rho_c = 1160; % Density of epoxy core (kg/m^3)
```

--- 2. Element Setup ---

```
eleLength = L / nElements;
DOF_per_node = 3; %
[u, w, phi] TotalDOF = nodes *
DOF_per_node;
```

```
K_global = zeros(TotalDOF,
TotalDOF); M_global =
zeros(TotalDOF, TotalDOF);
```

--- 3. Assemble Stiffness and Mass Matrices ---

```
for i =
    1:nElem
    ents n1
    = i;
    n2 = i+1;

    dof = [(n1-1)*3+1 : n1*3, (n2-1)*3+1 : n2*3];

    % Mid-element radius using axial
    coordinate x_mid = s(i) + eleLength/2;

    r_mid = R1 + x_mid*tan(alpha) % Axial stiffness EA
    EA = 2*E_f*h_face*2*pi*r_mid + E_c*h_core*2*pi*r_mid;

    % Bending stiffness EI
    d1 = h_core/2 + h_face/2;

    EI = E_f*2*pi*r_mid*(h_face^3/6 + 2*h_face*d1^2) ...
        + E_c*2*pi*r_mid*h_core^3/12;

    % Mass per unit length
    mu = 2*pi*r_mid*(2*rho_f*h_face + rho_c*h_core);

    % Local stiffness matrix
    k_ele = [EA/eleLength, 0, 0, -EA/eleLength, 0, 0;
            0, 12*EI/eleLength^3, 6*EI/eleLength^2, 0, -
            12*EI/eleLength^3, 6*EI/eleLength^2;
            0, 6*EI/eleLength^2, 4*EI/eleLength, 0, -6*EI/eleLength^2,
            2*EI/eleLength;
            -EA/eleLength, 0, 0, EA/eleLength, 0, 0;
            0, -12*EI/eleLength^3, -6*EI/eleLength^2, 0,
            12*EI/eleLength^3,
            -6*EI/eleLength^2;
            0, 6*EI/eleLength^2, 2*EI/eleLength, 0, -6*EI/eleLength^2,
            4*EI/eleLength];

    % Local mass matrix
    m_ele = (mu*eleLength/420)
    * ... [140, 0, 0,
    70, 0, 0;
    0, 156, 22*eleLength, 0, 54, -13*eleLength;
    0, 22*eleLength, 4*eleLength^2, 0, 13*eleLength, -
    3*eleLength^2;
    70, 0, 0, 140, 0, 0;
    0, 54, 13*eleLength, 0, 156, -22*eleLength;
    0, -13*eleLength, -3*eleLength^2, 0, -
    22*eleLength, 4*eleLength^2];
```

```
% Transformation matrix for conical angle
T = [cos(alpha), -sin(alpha), 0, 0, 0, 0;
     sin(alpha),  cos(alpha), 0, 0,
                                     0, 0;
     0, 0, 1, 0, 0, 0;
     0, 0, 0, cos(alpha), -sin(alpha), 0;
     0, 0, 0, sin(alpha),  cos(alpha), 0;
     0, 0, 0, 0, 0, 1];

k_global_ele = T * k_ele * T';
m_global_ele = T * m_ele * T';

K_global(dof, dof) = K_global(dof, dof) +
k_global_ele; M_global(dof, dof) = M_global(dof,
dof) + m_global_ele;
end
```

--- 4. Boundary Conditions ---

```
fixedDOF = [1, 2, 3]; % Clamped at
small end freeDOF = setdiff(1:TotalDOF,
fixedDOF);
```

--- 5. Solve Eigenvalue Problem ---

```
[V, D] = eig(K_global(freeDOF, freeDOF), M_global(freeDOF, freeDOF));
natural_frequencies = sqrt(diag(D))/(2*pi);
natural_frequencies = sort(real(natural_frequencies));
```

```
disp('First 5 Natural Frequencies (Hz):');
disp(natural_frequencies(1:5));
```

First 5 Natural Frequencies (Hz):

1.0e+04 *

0.0802

0.5993

0.6418

1.7674

2.2646

--- Plot Natural Frequencies ---

```
mode_number
s = 1:5;
figure;
plot(mode_numbers, natural_frequencies(1:5), '-o', ...
'LineWidth',2,'MarkerSize',8);

xlabel('Mode Number');
ylabel('Natural Frequency
(Hz)');
title('Natural Frequencies of CFRP-Epoxy Truncated Conical
Shell'); grid on;
set(gcf,'color','white')
```

References

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